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Interpretable Joint Movement Classification

via Combinatorial DTW Medoid Distances and
Ensemble Feature Selection

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Clinical Dilemma in Rehabilitation

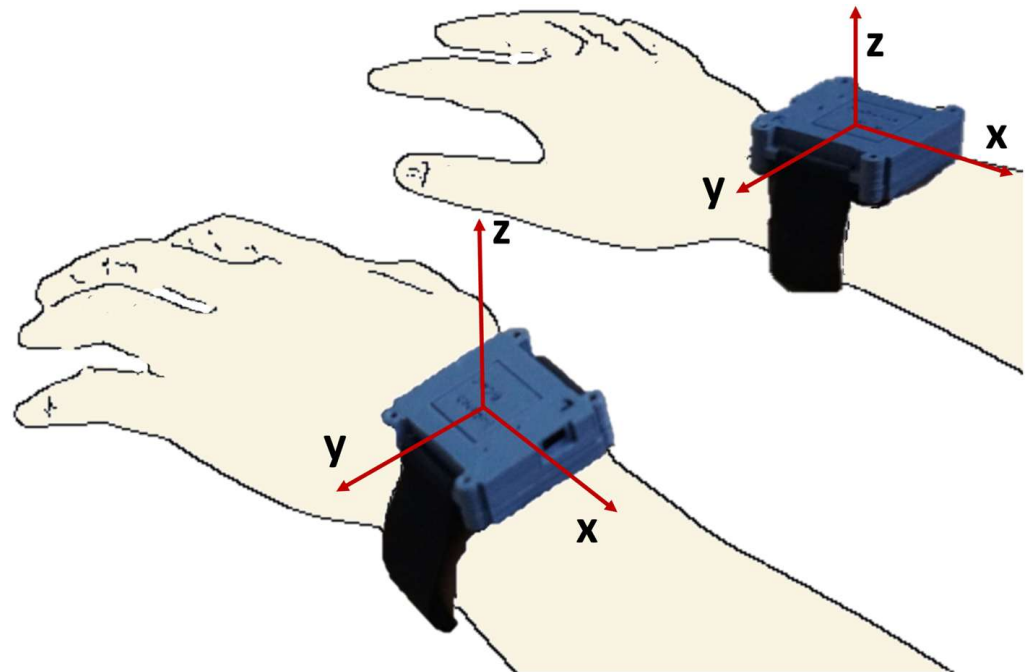
Single-Sensor Wrist Tracking

Deploying a single wrist-worn IMU is optimal for patient compliance in remote therapy. However, wrist, elbow, and shoulder movements generate overlapping kinematic signals at the wrist.

Limitations of Existing Methods

Single-Channel Limitations: Standard single-channel or basic X,Y,Z signals miss multi-axis coordination across joint movements.

High On-Device Cost: Running high-dimensional time-series analytics on smartwatches causes severe battery drain and memory overhead.

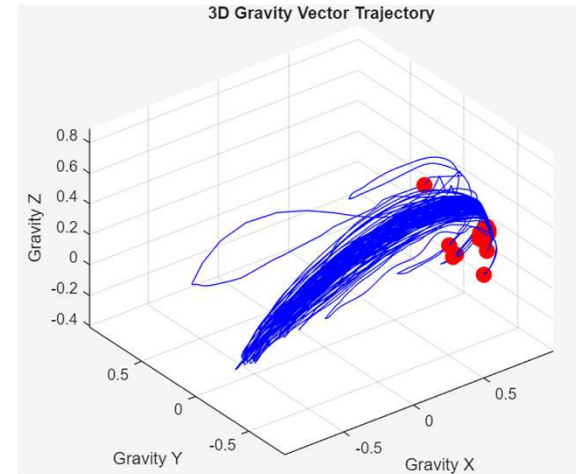


Cycles & DTW Medoid Baseline



Our Dataset

- The HemiPhysioData App is a WatchOS app that runs on Apple Watch.
- Data were collected from three adult participants
- Type of movement performed – elbow extension/flexion (el-exfl), shoulder extension/flexion (sh-exfl), and wrist pronation/ supination (wr-prsu)



Cycle Detection

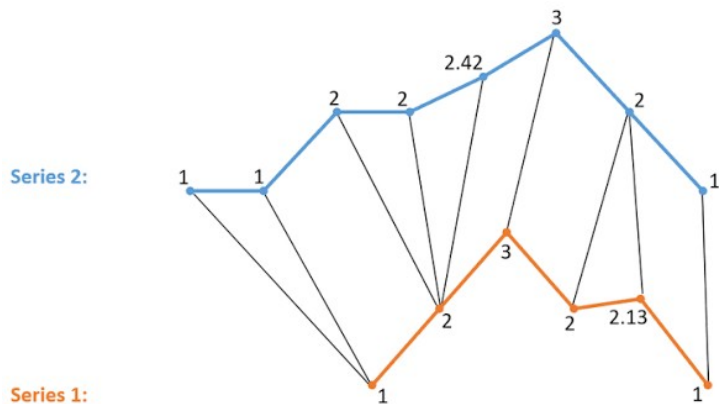
Autocorrelation segments continuous IMU streams into complete, biomechanically meaningful repetitions not $T = 100$ to samples.

Cycles & DTW Medoid Baseline



Dynamic Time Warping

Multivariate DTW nonlinearly aligns movement repetitions executed at varying speeds without phase distortion, preserving temporal dynamics.



Class Medoids

Instead of synthetic centroids (which produce unphysical average trajectories), medoids are real recorded cycles minimizing intra-class DTW distance for clinical replay.

Combinatorial Feature Space

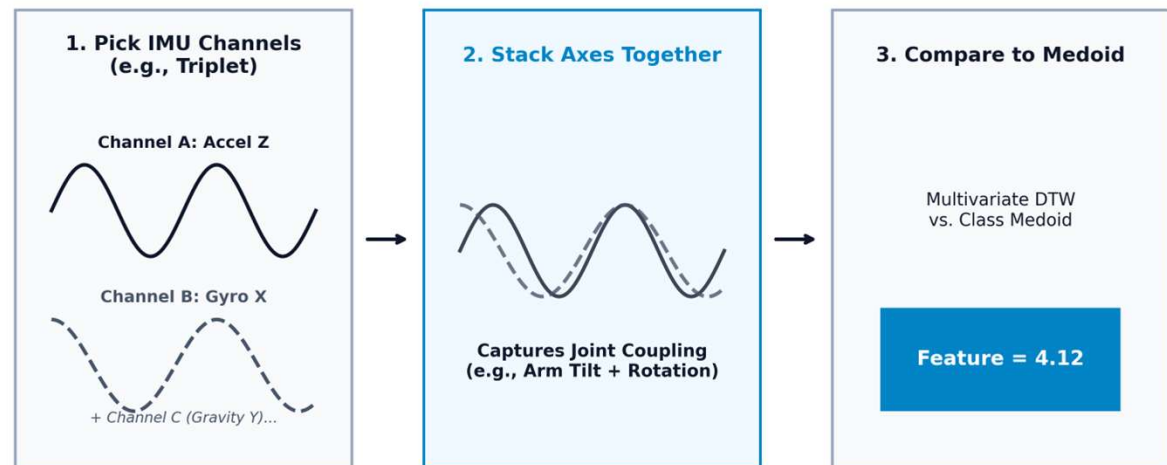
Multi-Axis Coupling

Upper limb motions engage multiple sensor channel simultaneously (e.g., rotation coupled with gravitatic acceleration shifts).

From 19 IMU channels, structured sub-channel com are constructed:

- **Single Channels:** $\binom{19}{1} = 19$ Subsets
- **Channel Pairs:** $\binom{19}{2} = 171$ Subsets
- **Channel Triplets:** $\binom{19}{3} = 969$ Subsets

Combinatorial Sub-Channel Feature Generation



3,477 Features

1,159 sub-channels × 3 class medoids

| Ensemble Feature Selection

75

Optimal Selected Features

<i>N</i>	Overall	Elbow	Shoulder	Wrist
50	91.01%	91.47%	84.35%	98.02%
75★	91.88%	90.70%	89.57%	96.04%
100	90.43%	92.25%	82.61%	97.03%
200	87.83%	87.60%	80.00%	97.03%
300	87.25%	87.60%	77.39%	98.02%
400	87.25%	90.70%	85.22%	85.15%
500	87.54%	88.37%	78.26%	97.03%
All (3,477)	88.41%	87.60%	80.00%	99.01%

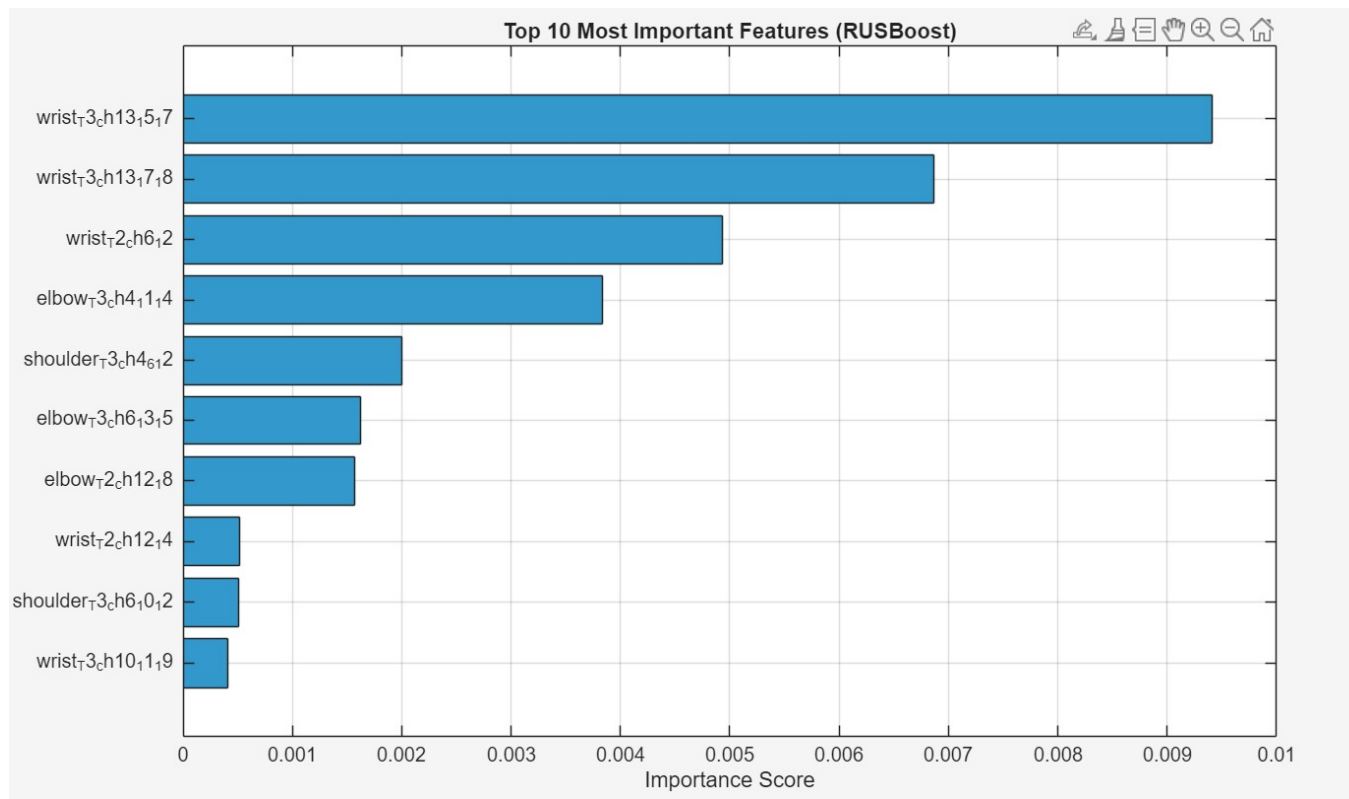
RUSBoost & "Less is More"

Addressing Redundancy: Evaluating all 3,477 combinatorial DTW features introduces noise and severe multicollinearity.

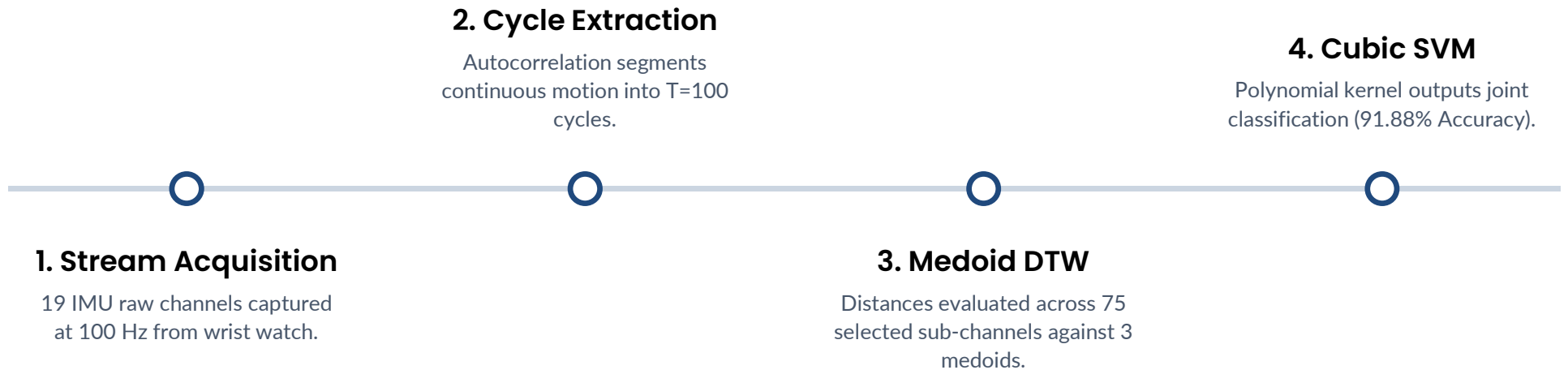
Class Imbalance: RUSBoost embeds random undersampling into boosting iterations to handle unequal cycle counts and rank predictor importance.

Performance Peak: Classification accuracy peaks sharply at **N = 75** (91.88%). Retaining all 3,477 features drops accuracy to 88.41% (>3% lower).

Ensemble Feature Selection



| End-to-End Execution Pipeline



Classifier Selection & Parameter Setup

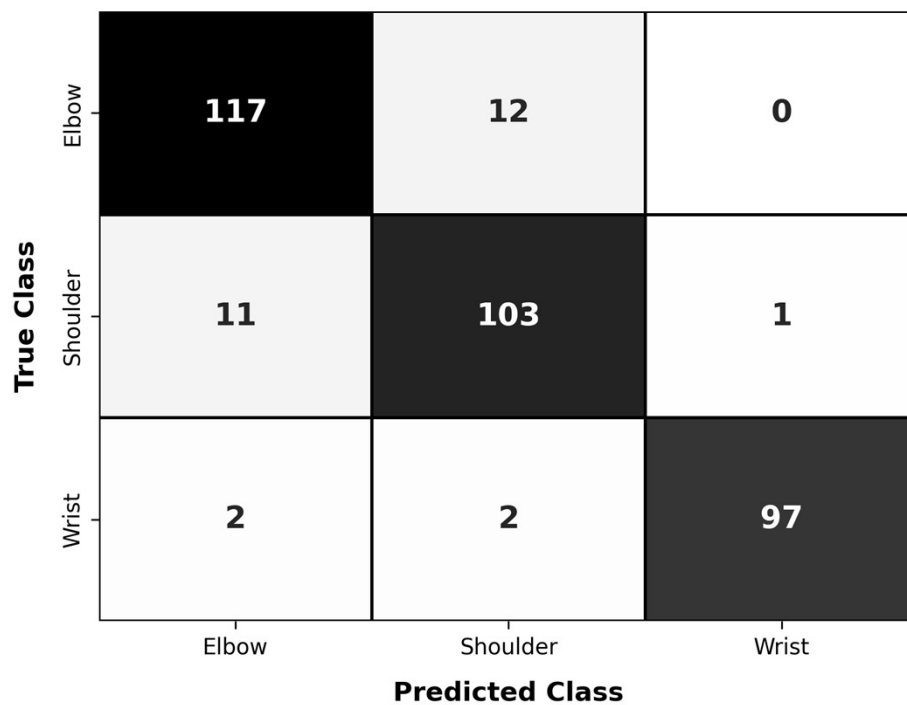
Kernel Choice (Cubic Kernel): Polynomial kernel of degree 3, $K(x, y) = (\gamma x^T y + r)^3$, models non-linear cross-channel interactions in DTW distance space.

Hyperparameter Configuration: Standardized features, Box Constraint $C = 1.0$, and auto-scaled kernel scale γ to balance decision margin and prevent overfitting.

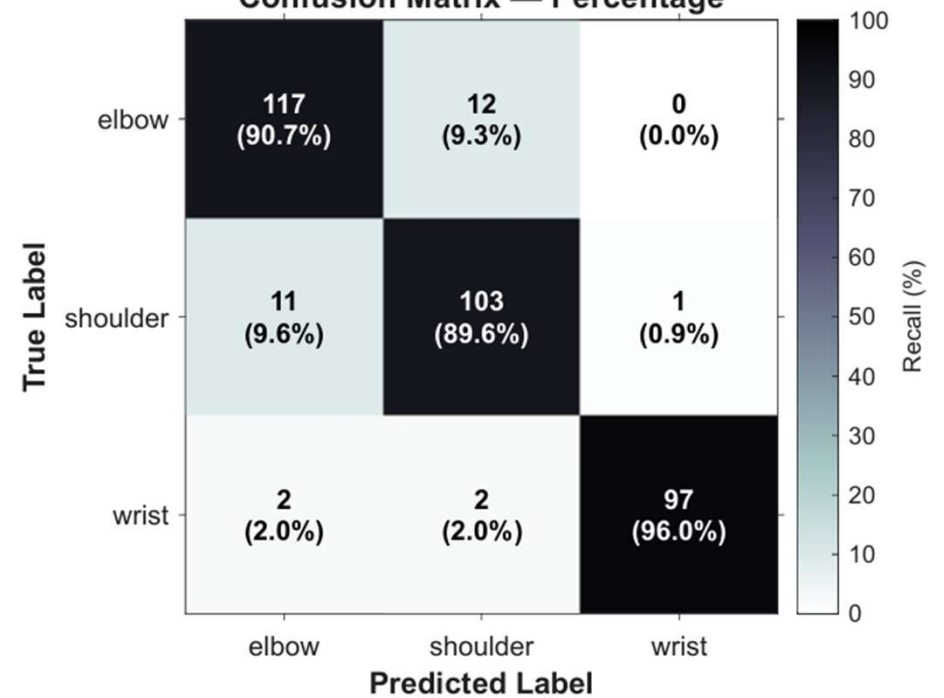
Multiclass Handling: One-vs-One (OVO) voting scheme to effectively resolve three-class decision boundaries (Wrist, Elbow, Shoulder).

Results

Test Set Confusion Matrix



Confusion Matrix — Percentage



Diagnostic Metrics & Coupling

Joint Class	Accuracy	Recall	ROC-AUC
Wrist	96.04%	0.998	0.991
Elbow	86.84%	0.981	0.624
Shoulder	89.57%	0.982	0.865
Macro Average	91.9%	92.1%	0.987

Strong Global Separability: Near-perfect macro ROC-AUC (0.987) confirms that 75 combinatorial DTW features effectively resolve cross-joint class boundaries.

High-Precision Wrist Isolation: Exceptional wrist performance (96.0% accuracy, 0.998 AUC) is driven by distinct rotational gyroscopic signals (pronation/supination).

Physiological Kinematic Coupling: Minor performance drop in elbow and shoulder classes reflects natural anatomical movement, where shoulder activation drags elbow trajectory.

Threshold Sensitivity: The high AUC relative to recall demonstrates robust class probability ranking across all operating points.

Benchmark Performance



Runtime Efficiency Advantage: While accuracy gains over 1-NN DTW are comparable, 1-NN DTW requires computing full 19-channel DTW against every training cycle. The proposed medoid pipeline yields a ~46-fold runtime reduction suitable for wearable edge deployment.

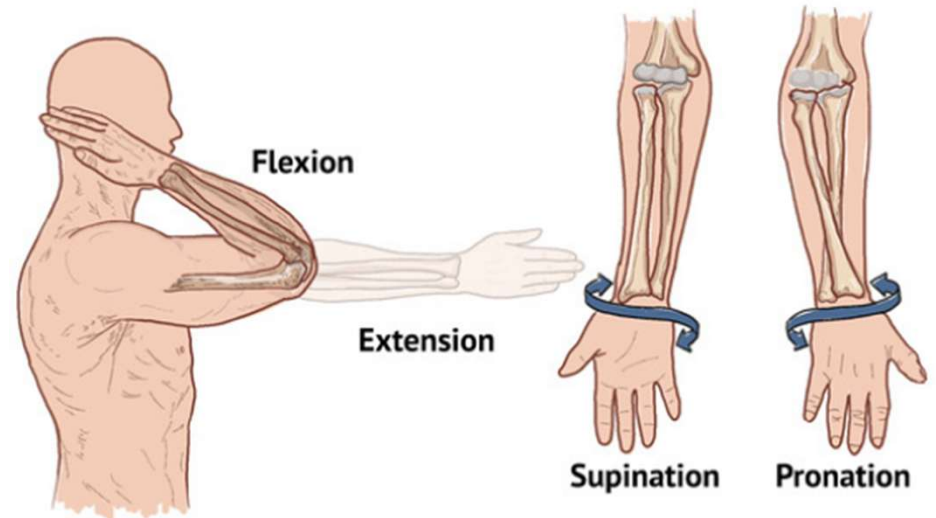
Clinical Impact & Future Scope

Clinical Impact

- ✓ **Physical Therapist Auditing:** Class medoids represent actual physical patient trials, allowing clinicians to visually replay and verify baseline templates.
- ✓ **Microcontroller Deployment:** Computing distances against only 3 static medoids drastically cuts RAM and battery draw for real-time wearables.

Future Scope

- ✓ Developing multi-joint prototype templates to resolve elbow-shoulder coupling.
- ✓ Validating the pipeline on larger cohorts using subject-independent cross-validation.
- ✓ Benchmarking real-time power draw and latency on wearable microcontrollers



Questions & Discussion

Thank you for your attention.

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